Deven Desai

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SUMMARY

Software Engineering Master's student and Stanford Hackathon winner specializing in reliable AI systems, evaluation infrastructure, and agentic workflows. Demonstrated track record at Parspec and Terminal Trends engineering full stack applications with Python, React, and TypeScript, building RAG pipelines that cut manual data extraction by 88 percent, and developing automated guardrails.

PROJECTS

OmniTrack AI | Agent Evaluation & Multimodal Data Processing Platform GitHub

- Processed unstructured streaming inputs and user metadata natively in cloud infrastructure by engineering a zero to one full stack data application using Python, React, and MongoDB.
- Achieved sub 100ms vector search latency by independently optimizing Qdrant indexing algorithms to support high concurrency, real time data retrieval queries.
- Improved system correctness and output reliability by 30 percent by building continuous evaluation loops, non deterministic guardrails, and automated verification suites.

AgenticFlow | Autonomous Execution Engine & Asynchronous Workflow Service GitHub

- Automated schema transformations and data migrations across heterogeneous cloud environments by architecting an asynchronous agentic execution engine using Python, Go, and LangGraph.
- Increased service execution throughput by 40 percent by designing concurrent reasoning loops and gRPC microservice interfaces as measured by system execution traces.
- Maintained strict data compliance and output auditability across platform boundaries by engineering a differential privacy validation layer verified via continuous automated testing.

High Performance Search Engine | Low Latency Data Discovery Infrastructure GitHub

- Analyzed and queried high volume structured datasets in real time by engineering a custom full stack analytical search interface as a solo developer.
- Decreased search query latency by 35 percent by developing optimized retrieval algorithms using Python and Java as measured by end to end response times.
- Optimized execution throughput by 20 percent by tuning underlying data structures and batch data execution routines.

WORK EXPERIENCE

Applied AI Intern, *Parspec, Inc., USA* **May 2026 – Dec 2026**

- Improved product-search relevance over a distributor inventory catalog, raising **recall@3 from 50% to 90%** on a **5,000-query** labeled evaluation set, by replacing keyword-only search with **hybrid** lexical and **semantic retrieval** fused via **reciprocal rank fusion**, re-scored with a **cross-encoder re-ranker**.
- Eliminated manual review of several-hundred-page construction documents for distributor bids, extracting structured product records at **88%** precision across **CSI Divisions 26 and 22**, by building a **RAG** pipeline over written specifications and schedules that grounds every extracted field in its source excerpt and auto-enriches contractor **Quantity Take-Off** line items.
- Automated lighting fixture product substitutions across 1.2M+ indexed datasheets at 85 percent match precision by engineering a retrieval pipeline that resolves free text inputs to anchor products and scores candidates via multi attribute filtering and semantic search.

Full Stack Developer, *Terminal Trends, India* **May 2024 – Dec 2024**

- Developed and optimized RESTful APIs using **Node.js** and **Express** to integrate real-time EV charging station data from multiple vendors (e.g., Tesla, government APIs), reducing server response time by **40%** and improving data reliability across the platform.
- Built interactive route-planning features with **React** and **Redux**, enabling users to locate nearby EV stations, view waiting times, pricing per kWh, and other live metrics along their journey, enhancing user convenience and engagement.

TECHNICAL SKILLS

AI Systems & Evals: Agentic Workflows (LangGraph), Multi Layered Guardrails, Non Deterministic Evals
Full Stack & Languages: Python, TypeScript, JavaScript, React, Redux, Node.js, Express, MongoDB, SQL, Go, C++, Java
Data Pipelines & Infrastructure: Distributed Systems, Microservices Architecture, gRPC, RESTful APIs, ETL Pipelines
Developer Tools & Practices: Git, CI CD, Automated Testing & Verification, Execution Tracing, System Design

EDUCATION

San Jose State University **San Jose, CA**
Master of Science in Software Engineering

ACHIEVEMENTS

- **Winner – Davis Wright Tremaine LLP Prize**, *Stanford Codex LLM x Law Hackathon (2025)*
  Led a 4-member team to build a **RAG-based legal QA engine** using Python, GPT-4, and vector-embedding retrieval with **cosine similarity**, enabling accurate extraction of contract clauses and delivering **90%+ correct answers**, outperforming 30+ teams.